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Retrospective evidence for clinical validity of expanded genetic model in warfarin dose optimization in a South Indian population

Aim: To optimize warfarin dose in patients at risk for thrombotic events, we have recently developed a pharmacogenomic algorithm, which explained 44.9% of the variability in warfarin dose requirements using age, gender, BMI, vitamin K intake, *CYP2C9* (*2 and *3) and *VKORC1* (*3, *4 and -1639 G>A) as predictors. The aim of the current study is to develop an expanded genetic model that can explain greater percentage of warfarin variability and that has clinical validity. **Patients & methods:** *CYP2C9**8, *CYP4F2* V433M, *GGCX* G8016A and thyroid status were added to an expanded genetic model (n = 243). **Results:** The expanded genetic model explained 61% of the variability in warfarin dose requirements, has a prediction accuracy of ± 11 mg/week and can differentiate warfarin sensitive and warfarin resistant groups efficiently (areas under receiver operating characteristic curves: 0.93 and 0.998, respectively; $p < 0.0001$). Higher percentage of International Normalized Ratios in therapeutic range (52.68 ± 4.21 vs 43.80 ± 2.27 ; $p = 0.04$) and prolonged time in therapeutic range (61.74 ± 3.18 vs 47.75 ± 5.77 ; $p = 0.03$) were observed in subjects with a prediction accuracy of < 1 mg/day compared with subjects with prediction accuracy > 1 mg/day. In the warfarin-resistant group, primary hypothyroidism was found to induce more resistance while in the warfarin-sensitive group, hyperthyroidism was found to increase sensitivity. **Conclusion:** The expanded genetic model explains greater variability in warfarin dose requirements and it prolongs time in therapeutic range and minimizes out-of-range International Normalized Ratios. Thyroid status also influences warfarin dose adjustments.

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KEYWORDS: anticoagulation ■ *CYP2C9* ■ International Normalized Ratio ■ INR ■ time in therapeutic range ■ TTR ■ *VKORC1* ■ warfarin

Warfarin is an oral anticoagulant widely used to prevent venous thromboembolism and systemic embolism in patients with prosthetic heart valves or atrial fibrillation [1]. Greater interindividual variability in warfarin dose requirements, narrow therapeutic range and adverse effects of inappropriate dose necessitated the need for clinical or genetic algorithms for dose optimization [2,3]. Clinical algorithms consider factors such as age, weight, height, comorbid conditions, concurrent medications and diet [4]. Genetic algorithms include polymorphisms in *CYP2C9* and *VKORC1* as additional predictors, as these two genes influence pharmacokinetics and pharmacodynamics of warfarin [5–9].

Among the different genetic algorithms developed, The International Warfarin Pharmacogenetics Consortium (IWPC) algorithm explained variability in dose requirements to a greater extent in the Caucasian population [10], however, in Asians its applicability was limited because of low frequency of *CYP2C9**2, *CYP2C9**3 and *VKORC1* -1639G>A variants [11]. In our recent study,

we proposed a population-specific pharmacogenomic algorithm by incorporating *CYP2C9**2, *CYP2C9**3, *VKORC1* -1639G>A, *VKORC1**3 and *VKORC1**4 genetic variants, along with age, gender, BMI and vitamin K intake thus explaining 44.9% ($r = 0.67\%$) of variability in warfarin dose requirement [11]. Certain genetic factors such as *CYP2C9**8 and *GGCX* G8016A are associated with warfarin sensitivity [12,13]; and *CYP4F2* rs2108622C>T variant associated with warfarin resistance [14]. This might explain additional variability to some extent. An expanded genetic model might improve the precision of pharmacogenomic prediction thus lowering out-of-range International Normalized Ratios (INRs) and prolonging the time in therapeutic range (TTR). With this hypothesis, we have developed an expanded genetic model by incorporating additional genetic variants, which are, *CYP2C9**8, *CYP4F2* V433M and *GGCX* G8016A, and also thyroid status from the data of 243 patients who were divided into derivation cohort (n = 160) and validation cohort (n = 83) based on response

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to warfarin therapy. Furthermore, this model was clinically validated retrospectively using TTR and number of INRs in therapeutic range as primary end points. The rationale to include thyroid status in the expanded genetic model is to explain warfarin sensitivity and warfarin resistance associated with hyperthyroidism and hypothyroidism, respectively.

Patients & methods

■ Study population & data collection

This was a retrospective study conducted on outpatients attending the department of cardiothoracic surgery, Nizam's Institute of Medical Sciences (NIMS), Hyderabad, India. The study was approved by the ethical committee of NIMS, Hyderabad, India (EC/NIMS/1170/2010). The indications for warfarin therapy were mainly for the prevention or treatment of thromboembolic diseases, such as patients with prosthetic-valve replacement, deep vein thrombosis or atrial fibrillation. The routine INR values were also recorded to determine the stable dose. A total of 243 subjects were segregated into derivation ($n = 160$) and validation cohorts ($n = 83$) based on two criteria: INR in therapeutic range (2–3 in aortic valve replacement [AVR]; 2.5–3.5 in mitral valve replacement [MVR]) at least three-times; and TTR >40%. In the derivation cohort, 84 males and 76 females were enrolled. The clinical presentation of these subjects is shown in Table 1. Out of 156 valve-replacement patients in the derivation cohort, five patients had MVR, two patients had AVR and MVR and rest of had AVR. NIMS has very few referrals for atrial fibrillation cases and hence limited number of cases were included in this study. Out of 160 subjects in the derivation cohort, 32 were warfarin sensitive, 115 were warfarin intermediate and 13 were warfarin resistant. In the validation cohort, 45 males and 38 females were enrolled. Out of the 83 valve-replacement patients in this cohort, three patients had MVR, one patient had AVR and MVR while the remainder had AVR. In two patients of the derivation cohort, bleeding events were reported during the warfarin initiation phase, which was stopped after attaining therapeutic INR, and one patient of the validation cohort had skin necrosis. Patient demographic details, medical history, medication used and daily warfarin dose were recorded. All the patients provided informed consent prior to enrollment in the study.

■ Genotyping

Blood samples (5 ml) were collected in ethylenediaminetetraacetic acid vacutainers. Total

DNA was extracted using standard protocols. Genotyping for *CYP2C9**2 (Arg144Cys; rs1799853), *CYP2C9**3 (Ile359Leu; rs1057910), *VKORC1* -1639G>A (rs9923231), *VKORC1**3 (G9041A; rs7294), *VKORC1**4 (C6009T; rs17708472), *VKORC1* D36Y (G106T; rs61742245), *CYP2C9**8 (c.449G>A;p.R150H; rs75838422), *CYP2C9**13 (c.269T>C; rs72558187), *CYP4F2* (V433M; rs2108622) and *GGCX* (G8016A; rs699664) was performed using the PCR-RFLP. All PCRs were carried out in a 25 µl volume containing 100 ng genomic DNA, 2.5 µl 10x PCR buffer (50 mM KCl, 10 mM Tris-HCl [pH: 9.0], 1.5 mM MgCl₂), 0.2 mM of deoxynucleotide triphosphates, 10 pM of each primers and 1 unit of Taq DNA polymerase.

For the *CYP4F2* V433M polymorphism, DNA was amplified using forward 5'-CGGAACCTTGACCACATCTACA-3' and reverse 5'-CCTACTCTCCCACAGGCA-TTA-3' primers. PCR reaction was comprised of initial denaturation at 95°C for 10 min followed by 35 cycles of 94°C for 30 s (denaturation), 60°C for 30 s (annealing) and 72°C for 1 min (extension) with a final extension step of 72 °C for 7 min. The PCR product was digested with 1 µl *PvuII* restriction enzyme. Restriction site is destroyed owing to the presence of variant allele.

The primers for *CYP2C9**8 and *CYP2C9**13 are the same, that is, forward 5'-TACA-AATACAATGAAAATATCATG-3', reverse 5'-CTAACAACCAGGACTCATAATG-3'. PCR conditions were: 95°C for 10 min followed by 35 cycles of 95°C for 30 s, 58°C for 30 s and 72°C for 45 s with a final extension step aof 72°C for 7 min. The PCR product was digested with 1 µl *Aci*I and 1 µl *Msp*I restriction enzyme to identify *CYP2C9**8 and *CYP29**13 variants, respectively. Loss of the *Aci*I restriction site suggests the presence of a *CYP2C9**8 variant. Creation of the *Msp*I restriction site suggests the presence of a *CYP2C9**13 variant allele.

For the *GGCX* G8016A polymorphism, PCR reaction was carried out using forward 5'-GCTGGTGTCTACTGCCCGC-3' and reverse 5'-GTACACTCCACAGCCCCTACAAA-3' primers. The reaction conditions were: 95°C for 10 min followed by 35 cycles of 94°C for 30 s, 59°C for 1 min and 72°C for 30 s with a final extension step of 72°C for 7 min. The PCR product was digested with 1 µl *Bst*UI restriction enzyme. Loss of restriction site suggests the presence of a *GGCX* 8016A allele. All the digested PCR products were visualized by agarose gel electrophoresis. The conditions

for the analysis of other polymorphisms are described elsewhere [11]. To validate the accuracy of genotyping using PCR-RFLP, each PCR reaction was accompanied by a negative control (no DNA) and a positive control (known heterozygous/homozygous mutant). Each lot of restriction enzyme was checked for its activity using plasmids with known restriction sites. Approximately 20% of the samples were re-analyzed and 100% concordance in genotyping was obtained.

Biochemical analysis

In both derivation and validation cohort subjects, we have analyzed plasma thyroid-stimulating hormone levels using thyroid stimulating hormone ELISA Micro plate Test (Calbiotech, Inc.) and plasma thyroxine (T4) levels using T4 ELISA kit (Weldon Biotech Pvt. Ltd). The interassay and intra-assay precision were 4.1 and 4.7% respectively. Using these two markers coupled with clinical details, subjects were categorized into four groups: euthyroid, primary hypothyroidism, secondary hypothyroidism and hyperthyroidism. In patients from the derivation cohort, out of 32 warfarin-sensitive patients, seven had primary hypothyroidism, three had secondary hypothyroidism, two had hyperthyroidism and the rest were euthyroid. Out of 115 patients with intermediate warfarin requirement, nine had primary hypothyroidism, five had secondary hypothyroidism, eight had hyperthyroidism and the rest were euthyroid. In the warfarin resistant group (n = 13), two had primary hypothyroidism, two had secondary hypothyroidism and two had hyperthyroidism and the rest were euthyroid. Except four cases of primary hypothyroidism who were already receiving eltroxin treatment, all other cases were freshly diagnosed during the course of the study.

Statistical analysis

All the SNPs were tested for deviation from Hardy-Weinberg equilibrium using χ^2 test between the observed and expected frequencies. Student's t-test was used to assess statistically significant differences among the continuous

Table 1. Distribution of demographic and genetic parameters across derivation and validation cohorts.

Parameter	Derivation cohort (n = 160)	Validation cohort (n = 83)	p-value
Age (years)	38.5 ± 15.1	36.9 ± 15.7	0.47
Male:female ratio	84:76	45:38	0.98
BMI (kg/m ²)	21.62 ± 4.25	21.29 ± 4.35	0.58
Clinical implication			
Valve replacement	156	83	0.37
Deep vein thrombosis	2	0	0.78
Atrial fibrillation	2	0	0.78
Genotype			
CYP2C9*2	28/320 (8.75%)	19/166 (11.44%)	0.43
CYP2C9*3	30/320 (9.37%)	14/166 (8.43%)	0.86
VKORC1 -1639 G>A	43/320 (13.44%)	17/166 (10.24%)	0.38
VKORC1*3	58/320 (18.12%)	18/166 (10.84%)	0.05
VKORC1*4	28/320 (8.75%)	9/166 (5.42%)	0.26
CYP4F2 V433M	165/320 (51.56%)	86/166 (51.80%)	1.00
GGCX G8016A	30/320 (9.37%)	22/166 (13.25%)	0.25
CYP2C9*8	5/320 (1.56%)	4/166 (2.41%)	0.76

For age and BMI, p-value was obtained by student t-test and data is represented as mean ± standard deviation. Fisher's exact test was used for other variables. Genotype data is represented as variant allele to total alleles ratio indicating variant allele frequency.

variables of the derivation and validation cohorts. Data on the derivation cohort was segregated according to gender and two multiple linear regression models were developed. For the regression models, age, BMI, CYP2C9*2, CYP2C9*3, CYP2C9*8, VKORC1*3, VKORC1*4, VKORC1 -1639G>A, GGCX G8016A, CYP4F2 V433M and thyroid status were used as the predictor variables and therapeutic dose as the outcome variable. The predicted dose was correlated with therapeutic dose using linear regression analysis. Bland and Altman analysis was used to assess the agreement between predicted and therapeutic doses of warfarin. The discriminatory potential of developed algorithm was tested using receiver operating characteristic (ROC) curves with two cutoff points of 21 and 49 mg/week. The area under the receiver operating curve (AUC) was used as an indicator of discriminatory potential. The applicability of the new algorithm versus

Box 1. Algorithms were proposed for warfarin dose prediction.

- Males = $-(0.1013885349 \times \text{age}) + (1.449999606 \times \text{BMI}) + (8.054730665 \times \text{CYP2C9*2}) + (1.726919455 \times \text{CYP2C9*3}) - (4.437335987 \times \text{VKORC1*3}) - (2.771903482 \times \text{VKORC1*4}) + (1.511628517 \times \text{VKORC1 -1639}) + (1.570215716 \times \text{CYP4F2 V433M}) + (2.409742997 \times \text{GGCX}) + (11.05198035 \times \text{CYP2C9*8}) + 7.970140851$
- Females = $-(0.05440552061 \times \text{age}) - (0.2938201651 \times \text{BMI}) - (1.576151039 \times \text{CYP2C9*2}) - (5.950436495 \times \text{CYP2C9*3}) + (2.983528309 \times \text{VKORC1*3}) + (8.699010214 \times \text{VKORC1*4}) - (11.00733747 \times \text{VKORC1 -1639}) - (2.282918521 \times \text{CYP4F2 V433M}) - (4.097105716 \times \text{GGCX}) - (2.96671589 \times \text{CYP2C9*8}) + 44.53497515$

Table 2. Impact of thyroid status on interindividual variability of warfarin dose.

Variable	Euthyroidism	Hypothyroidism	p-value
Difference between therapeutic vs predicted dose (mg/week)			
Warfarin sensitive	-6.85 (5.33)	-6.56 (5.45)	0.92
Warfarin intermediate	0.32 (5.79)	2.76 (6.37)	0.14
Warfarin resistant	8.11 (5.96)	16.77 (3.44)	0.045*
End points to test clinical validity			
INRs in therapeutic range (%)	54.87 (22.86)	43.73 (18.23)	0.024*
TTR (%)	58.85 (24.25)	47.19 (17.10)	0.029*

All data is represented as mean $\pm$ standard deviation.

*p-value is statistically significant.

INR: International Normalized Ratio; TTR: Time in therapeutic range.

IWPC was tested across wide range of warfarin dose requirements. The applicability of the algorithm in predicting optimal dose was tested in the validation cohort. Using epimax calculator, sensitivity, specificity, rate of over estimation and rate of underestimation were calculated. All other analyses were performed using GraphPad Prism (version 4.00 for Windows). A p-value <0.05 was considered to be statistically significant.

Results

All the genetic polymorphisms were in accordance with Hardy–Weinberg equilibrium. As shown in TABLE 1, no statistically significant differences were observed in the distribution of demographic and genetic parameters between the derivation and

validation cohorts (TABLE 1). Two polymorphisms, that is, *VKORC1* D36Y and *CYP2C9*13* were absent in both the groups and hence excluded from the expanded genetic model.

Based on the expanded genetic model, following pharmacogenomic algorithms were proposed for warfarin dose prediction (Box 1).

Furthermore, the predicted doses of warfarin were adjusted according to thyroid status using the information provided in TABLE 2.

Demographic characteristics, specifically age, BMI and gender, explained 12.93% of the variability in warfarin dose requirements; addition of the *CYP2C9*2*, *CYP2C9*3*, *VKORC1*3*, *VKORC1*4*, *VKORC1* -1639G>A genes to the model explained 44.9% of the variability in warfarin dose requirements. *CYP4F2* V433M, *GGCX* G8016A, *CYP2C9*8* variants explained 0.35, 0.01 and 5.1% variability, respectively. Incorporation of thyroid status to this model explained 10.44% variability in warfarin dose requirements (FIGURE 1); however, the univariate model of thyroid status explained 2.5% variability in females and 1.69% variability in males.

As shown in FIGURE 2, Bland and Altman analysis between the predicted versus therapeutic dose (in the derivation cohort) validated the applicability of this algorithm by showing good agreement between both the methods (bias: 0.0033 mg/week; 95% limits of agreement: ± 11 mg/week).

Segregating the derivation cohort based on therapeutic dose of warfarin into warfarin sensitive (<21 mg/week), warfarin intermediate (22–49 mg/week) and warfarin resistant (>49 mg/week) groups helped in evaluating the prediction accuracy of the developed algorithm over a wide range of warfarin doses (FIGURE 3).

By using 21 mg/week and 49 mg/week as cutoff points, the applicability of the expanded genetic model was tested by plotting ROC

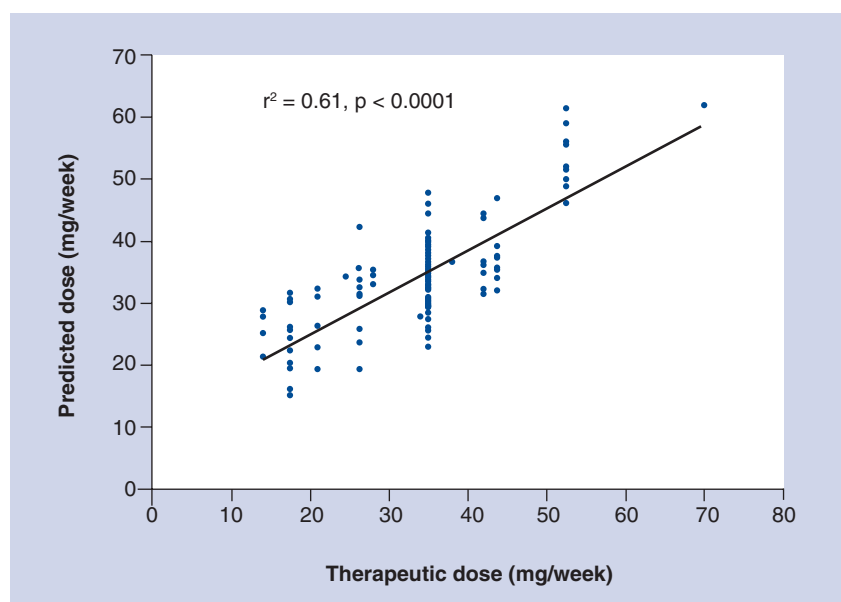


Figure 1. Linear regression analysis of therapeutic versus predicted dose. X-axis represents therapeutic dose; Y-axis represents predicted dose (in mg/week) of warfarin. The linear regression analysis showed good correlation between therapeutic and predicted doses ($r^2 = 0.61$, $p < 0.0001$).

curves. The current model showed 93% accuracy in differentiating <21 mg/week and >21 mg/week (FIGURE 4A); and 99.8% accuracy in differentiating <49 mg/week and >49 mg/week (FIGURE 4B).

As shown in FIGURE 5A & FIGURE 5B, higher percentage of INRs in therapeutic range (52.68 ± 4.21 vs 43.80 ± 2.27 ; $p = 0.04$) and prolonged TTR (61.74 ± 3.18 vs 47.75 ± 5.77 ; $p = 0.03$) were observed in subjects with prediction accuracy of <1 mg/day compared with subjects with prediction accuracy >1 mg/day.

Compared to standard empirical dosing, the pharmacogenomic predicted dose showed greater accuracy (89 vs 59.8%), greater sensitivity (89 vs 58%), greater specificity (89.7 vs 68.6%), lower rate of overestimation (10.3 vs 31.3%) and lower rate of underestimation (11.2 vs 42%).

As shown in FIGURE 6, in warfarin resistant individuals, primary hypothyroidism was shown to further induce resistance. In warfarin sensitive individuals, hyperthyroidism was shown to further induce sensitivity.

Discussion

The contribution of demographic characteristics such as age, gender and BMI to interindividual variability in warfarin dose requirements is independent of ethnicity with all the existing studies showing a similar trend [15]. Among the demographic characteristics, younger age, male gender and high BMI require higher warfarin dose [16,17]. The contribution of genetic factors to interindividual variability depends on the distribution of these variants in a particular

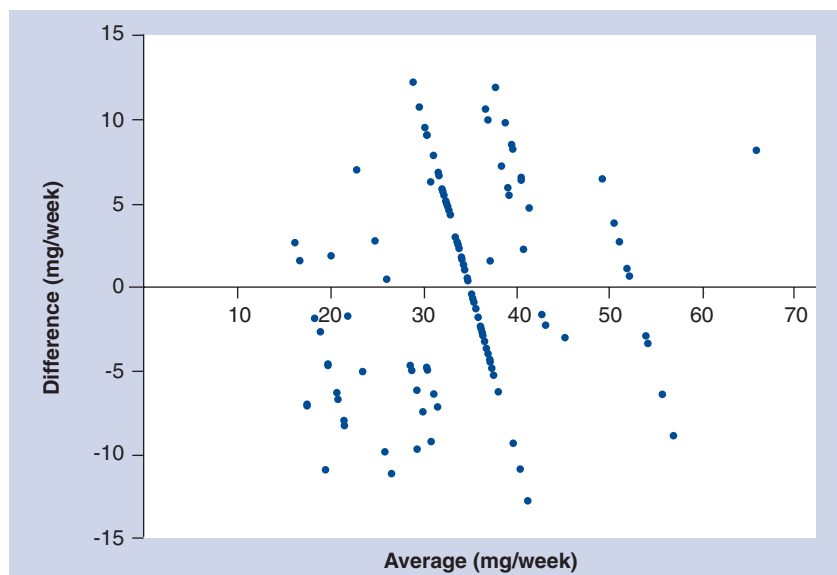


Figure 2. Bland and Altman analysis of therapeutic versus predicted dose.

X-axis represents average of therapeutic and predicted dose; Y-axis represents difference between therapeutic and predicted doses. The Bland and Altman analysis showed a bias of 0.0033 with 95% limits of agreement being -11 to +11 mg/week.

ethnic group or population [18,19]. This explains the limited applicability of IWPC algorithm in the Indian population. The IWPC algorithm was validated for patients with INRs between 2–3 and hence might not explain dose variability for INRs between 3–3.5. Another reason for limited applicability of IWPC could be owing to different genetic constitution of Indians compared with other populations such as Caucasians and Jews and so on [20,21]. The major differences in the genetic constitution of Indians compared with other populations are: the *VKORC1* D36Y polymorphism that has a

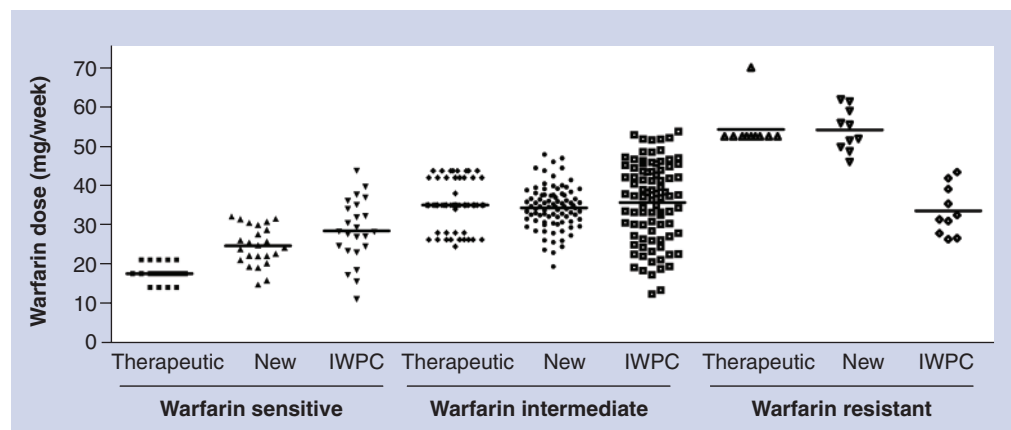


Figure 3. Comparison of predicted warfarin dose with International Warfarin

Pharmacogenetics Consortium predicted dose. The therapeutic and predicted (using current method and IWPC) doses of warfarin were compared across three categories, that is, warfarin sensitive, warfarin intermediate and warfarin resistant. There is close proximity between therapeutic dose and predicted dose by the current method compared with IWPC predicted dose. IWPC: International Warfarin Pharmacogenetics Consortium.

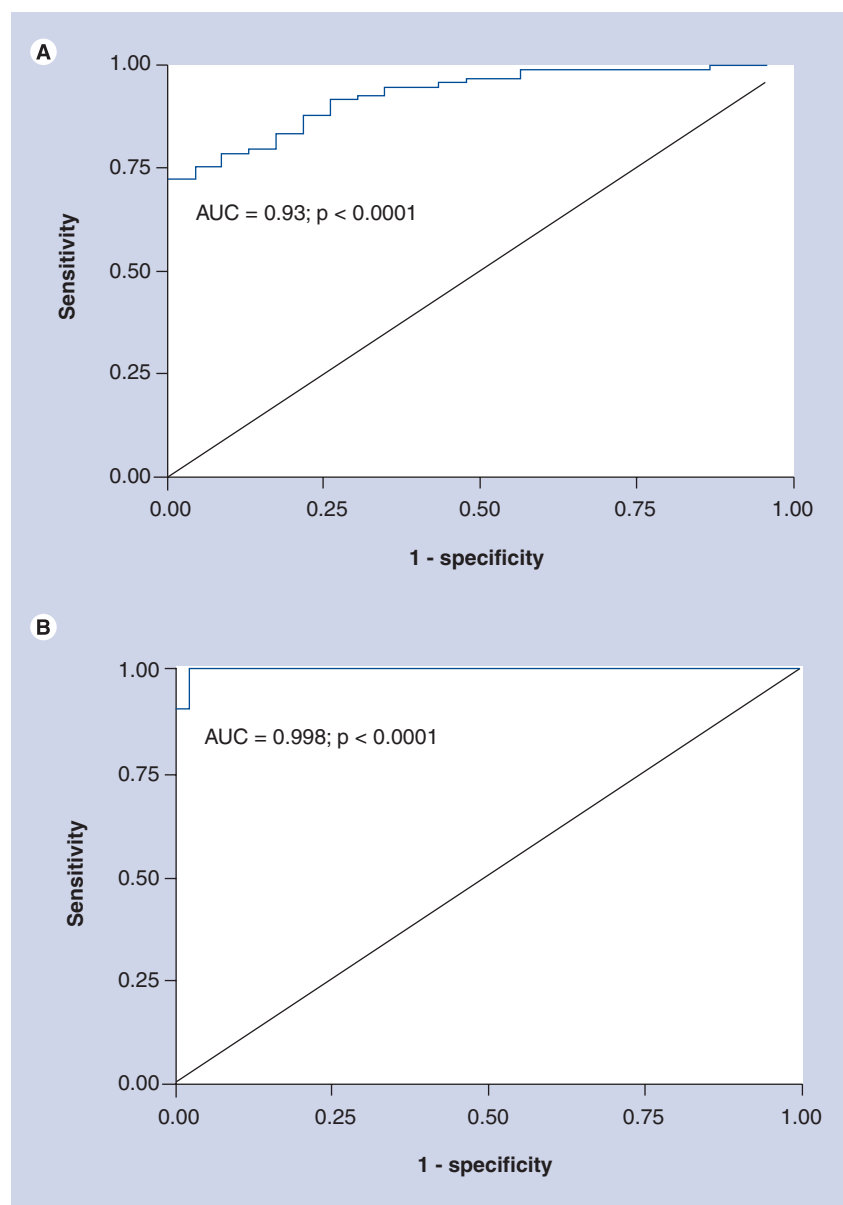


Figure 4. Receiver operating characteristic curves: as a measure of discriminatory potential. (A) Receiver operating characteristic curve plotted with predicted values to test the utility of prediction in distinguishing <21 mg/week and >21 mg/week. This plot shows an area under the curve of 0.93 indicating good discriminatory potential. (B) Receiver operating characteristic curve was plotted to test the utility of prediction in distinguishing <49 mg/week and >49 mg/week. This plot shows an area under the curve of 0.998 indicating good discriminatory potential.

AUC: Area under the receiver operating curve.

prevalence of 15% in Ethiopian Jews and 4% in Ashkenazi Jews, is absent in Indians; the *CYP2C9*13* polymorphism that has a prevalence of 0.7% in the Han Chinese population, is absent in Indians [22,23]; and a relatively low frequency of homozygous mutant genotypes of *CYP2C9*2* and *CYP2C9*3* in Indians [17]. A study by Chan *et al.* showed contribution of the *CYP4F2* V433M polymorphism in explaining a greater extent of interindividual variability

in warfarin dose requirement in other Asian populations [24]. Despite the high prevalence of *CYP4F2* V433M in our population, its contribution to interindividual variability was limited. This highlights that among Asians there are also wide ethnic and population-level variations in the frequencies of genetic polymorphisms in genes regulating warfarin metabolism. These observations necessitate the need for population-specific algorithms. The limited utility of *CYP4F2* V433M in warfarin dose optimization as observed in the current study was consistent with the Shahin *et al.* study [25].

Our earlier algorithm using age, gender, BMI, vitamin K intake, *CYP2C9*2*, *CYP2C9*3*, *VKORC1* -1639G>A, *VKORC1*3* and *VKORC1*4* explained 44.9% of the variability in warfarin dose requirements among Indians [11]. The current expanded genetic model incorporating *CYP4F2* V433M, *GGCX* G8016A, *CYP2C9*8* and thyroid status increases this variability by 16.11%. This study shows association of hyperthyroidism and hypothyroidism with warfarin sensitivity and warfarin resistance, respectively, consistent with the previous studies [26–28]. In hyperthyroidism, there is an increase in the rate of metabolism of certain coagulation factors and decrease in plasma-protein binding of warfarin, which contributes to lower warfarin dose requirements [29]. On the contrary, hypothyroidism results in a 50% decrease of the metabolic rates of prothrombin, factor VII and factor X and hence show resistance to warfarin [30]. Incorporation of thyroid status to the genetic models improves precision in warfarin dose prediction as shown in the current study.

The Bland and Altman analysis, using therapeutic dose as the gold standard showed good limits of agreement with predicted dose, thus validating the developed algorithm. When the therapeutic dose was compared with the predicted dose according to the current algorithm versus the IWPC predicted dose, the current algorithm fared well in warfarin sensitive and warfarin resistant groups. In the intermediate range, the predicted values by both the methods are equally compatible.

The major concern for the treating physician is to know whether a patient is warfarin sensitive, warfarin intermediate or warfarin resistant. To address the issue of clinical application of the expanded genetic model, ROC curves were plotted for our data, which demonstrated good potential in differentiating the subjects of these three subgroups.

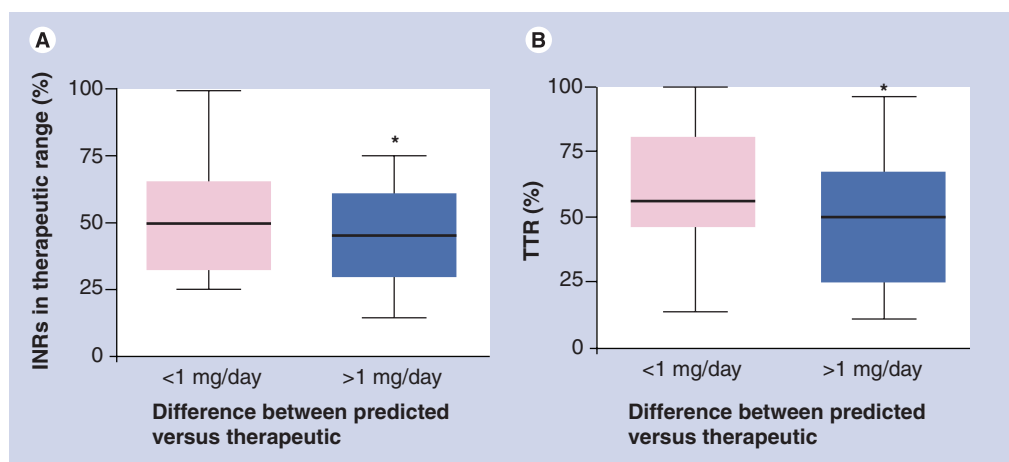


Figure 5. Retrospective analysis to justify the clinical validity of pharmacogenomic prediction in patient management. The data on derivation cohort was categorized into two subgroups: prediction accuracy <1 mg/day and prediction accuracy >1 mg/day. Using these categories the number of INRs in therapeutic range (A) and time in therapeutic range (B) were compared using student t-test. In subjects with a more precise prediction, a higher number of INRs in therapeutic range ($p = 0.03$) and a prolonged time in therapeutic range ($p = 0.04$) was observed compared with the subjects with a prediction accuracy of >1 mg/day.

*Statistically significant p-value.

INR: International Normalized Ratio; TTR: Time in therapeutic range.

The clinical validity of such algorithms depends on how this predicted dose will be useful in lowering out-of-range INRs and how they prolong the time in the therapeutic range. Gong *et al.* and Anderson *et al.* observed 64.8 and 69.7% TTR in pharmacogenetic guided

groups [31,32]. Although our study is not prospective in nature, we evaluated the validity of pharmacogenomic algorithm retrospectively by classifying the subjects in therapeutic range into two groups: prediction accuracy <1 mg/day and >1 mg/day. We observed that in subjects

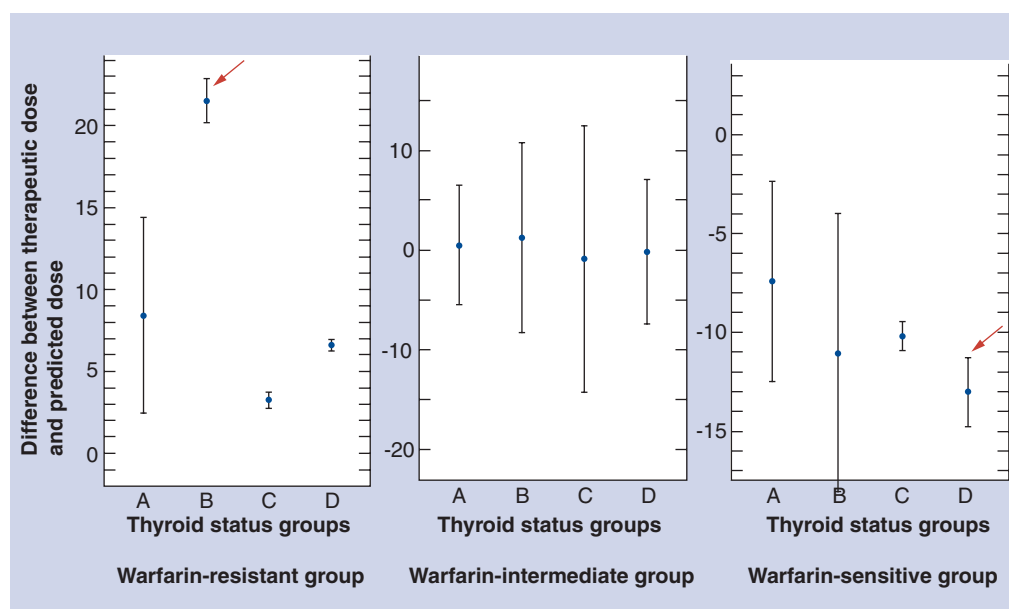


Figure 6. Impact of thyroid status on warfarin dose requirement. Patients were segregated into warfarin sensitive (≤ 21 mg/week), warfarin intermediate (22–49 mg/week) and warfarin resistant (> 49 mg/week). The difference between therapeutic dose and predicted dose was plotted with respect to thyroid status: A: euthyroid; B: primary hypothyroidism; C: secondary hypothyroidism; D: hyperthyroidism. In subjects with warfarin resistance, primary hypothyroidism showed further resistance. In subjects with warfarin sensitivity, hyperthyroidism showed further sensitivity. Arrows represent major deviation from rest of the groups. Primary hypothyroidism is strongly associated with warfarin resistance while hyperthyroidism is strongly associated with warfarin sensitivity.

with prediction accuracy <1 mg/day, the TTR is prolonged and out-of-range INRs are lower suggesting that such algorithms might benefit the patients in prevention of adverse events as evidenced by reduced rates of underestimation and overestimation.

In the current study, most of the enrolled subjects were very young and were being treated with warfarin following heart valve replacement, which could be due to high incidence of valvular disease in India compared with its western counterparts and predominant in younger age groups [33]. Furthermore, patients undergoing mechanical valve replacement are usually at a higher risk of thrombosis and therefore need greater anticoagulation activity than patients suffering from atrial fibrillation or deep vein thrombosis. To the best of our knowledge, this algorithm is the first expanded genetic model that incorporated thyroid status as an additional variable, and hence can benefit the patients undergoing warfarin therapy irrespective of thyroid status. The strengths of the current study are: explaining greater interindividual variability, reducing the rate of overestimation and underestimation of warfarin dose, indicating clinical benefits such as prolonged TTR and low frequency of out-of-range INRs with precise dose prediction. Although, our algorithm fared better in the warfarin-sensitive group; the dosing is still not very precise in this group. The PCR-RFLP technique, which was used for genotyping in this study, is laborious and chances of human errors are high when applicable to large studies of clinical relevance. A good quality check, as mentioned in the methodology, will ensure accuracy in genotyping. However, we are planning to adopt high-throughput screening methods for randomized trials so as

to minimize screening time. This study is retrospective in nature and hence a prospective randomized trial should be conducted to test the clinical utility of developed algorithm. Furthermore, end points such as number of adverse drug reactions and duration of hospitalization should be measured to test the clinical utility. Some of the polymorphisms showed very low frequency of variant alleles and hence future studies with relatively large sample sizes are warranted to corroborate our findings further [25,34,35].

Conclusion

The expanded genetic model can cater to the needs of warfarin dose optimization in over a wide range of warfarin dose requirements.

Future perspective

Expanded genetic models of additional variables can be useful in improving precise prediction of warfarin dose. The improved precision might reduce adverse effects associated with inappropriate dose of the drug and decrease the rate of hospitalization. Randomized clinical trial should be conducted to test the effectiveness of this model in patient care.

Financial & competing interests disclosure

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Executive summary

- *VKORC1* and *CYP2C9* are the major genes involved in the metabolism of warfarin for the prevention of thromboembolic diseases.
- However, greater interindividual variability, narrow therapeutic range and adverse effects of inappropriate dose necessitated the need for clinical or genetic algorithms for dose optimization.
- A total of 243 patients who were on warfarin therapy were genotyped to detect the *CYP2C9**2, *CYP2C9**3, *CYP2C9**8, *CYP2C9**13, *CYP4F2* V433M, *VKORC1**3, *VKORC1**4, *VKORC1* D36Y, *VKORC1* -1639 and *GGCX* G8016A polymorphisms.
- Demographic characteristics of age, BMI and gender explained 12.93% of the variation while addition of genetic factors explained 50.36% of the variability.
- The incorporation of thyroid status to the expanded genetic model contributed to 61% of the variation in warfarin dose requirements.
- Compared to standard empirical dosing, the pharmacogenomic predicted dose showed greater accuracy, sensitivity, specificity and lower rate of overestimation and underestimation.
- The current study explains greater interindividual variability, reduction in the rate of overestimation and underestimation of warfarin dose, indicates clinical benefits such as prolonged time in therapeutic range and low frequency of out-of-range International Normalized Ratios with precise dose prediction.

Ethical conduct of research

The authors state that they have obtained appropriate institutional review board approval or have followed the principles outlined in the Declaration of Helsinki for all human or animal experimental investigations. In addition, for investigations involving human subjects, informed consent has been obtained from the participants involved.

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